

UBT838: STEM CELL TECHNOLOGY

L	T	P	Cr
3	0	0	3.0

Course Objective: The objective of this course is to enable Students to understand the principles of stem cells. Students will acquire knowledge in the areas of tissue engineering.

Syllabus

Concept of Stem Cells: Stem cells: Basic concepts and properties, Totipotency, Pluripotency, Embryonic stem cells, Germinal stem cells, Adult stem cells, Tumor stem cells, Stem cell plasticity, General methods of characterization of stem cells.

Stem cell self-renewal and pluripotency: molecular mechanisms Cell cycle regulation in stem cells. Stem cell niches, Stem cell lineage tracing.

Embryonic stem (ES) cells: Isolation of ES cells, Salient features and application of ES cells, ES cells. Human and Mouse embryonic stem cells, Differentiation of ES cell, Maintenance of ES in undifferentiated state.

Hematopoietic Stem Cells (HSC): Identification and Characterization of HSCs, Sources of HSC Mouse Assay of HSC, HSC in leukemia and lymphoma, Clinical use of HSC.

Mesenchymal and Neural Stem Cell: Embryonic origin of MSC's, Harvesting, Isolation and Characterization, Differentiation studies of MSC's, Neural stem cell and Neural crest stem cell.

Stem Cells and Cloning: Therapeutic and reproductive cloning, Nuclear Transfer method, Application of NT ES cells, Safety of NT ES cells.

Application of stem Cells: Overview of embryonic and adult stem cells for therapy Neurodegenerative diseases; Parkinson's, Alzheimer, Spinal Cord Injuries and other brain Syndromes; Tissue system Failures; Diabetes; Cardiomyopathy; Kidney failure; Liver failure; Cancer; Hemophilia etc. Applications of stem cells in medicine and different disease models, Biosafety and Stem cell research, Regulatory considerations and FDA requirements for stem cell therapy.

Laboratory Work (if applicable)

Course Learning Objectives (CLO)

The students will be able to:

1. Comprehend the concept of stem cells, different types of stem cells.

Approved in 1st meeting of the Academic Council held on September 11, 2025. Further revision approved in 3rd meeting of the Academic Council held on March 18, 2026, incorporating modification in credits due to the addition of AI courses.

2. Describe the concept of stem cell cloning and its applications.
3. Recognize treatment of human diseases connected to stem cell therapy.

Text Books

1. Twyman RM, Developmental Biology Viva Books Pvt. Ltd. (2001)
2. Marshak L, Stem Cell Biology, Cold Spring Harbor Publication, (2001).
3. Lanza RP, Robert Langer R and Chick WL, Principles of Tissue Engineering, Academic Press (1997).

Reference Books

1. Palsson B and Bhatia S. Tissue Engineering, Pearson-Prentice Hall, (2003).

Evaluation Scheme:

Sr. No.	Evaluation Elements	Weightage (%)
1.	MST	25-35
2.	EST	35-45
3.	Sessionals (May include assignments/quizzes)	30

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